

What Factors Are Most Associated with Football Player Market Value?

DATA 400 Individual Mini-Project

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Question and motivation

RESEARCH QUESTION

Which player characteristics and performance statistics are most strongly associated with a football player's market value?

- Player valuations vary enormously, from tens of thousands to hundreds of millions of euros.
- Players with similar output can carry very different values.
- Goal: describe which characteristics move with value, not predict a "true price".

Market value $\neq$ transfer fee

Transfermarkt's estimate at a point in time: not a price paid, not a true value.

WHO CARES

Fans and journalists who quote values, analysts and recruiters who use them, players and agents affected by them.

Data: five seasons of Transfermarkt history

32,046

player-seasons

13,511

distinct players

14

domestic European
leagues

5

seasons, 2020/21 to
2024/25

One row = one player in one season

- Source: David Caribou's public transfermarkt-datasets (CC0)
- Match records summed into season totals
- A player appears once for each season played

Main variables

Age

Position

Minutes

Goals

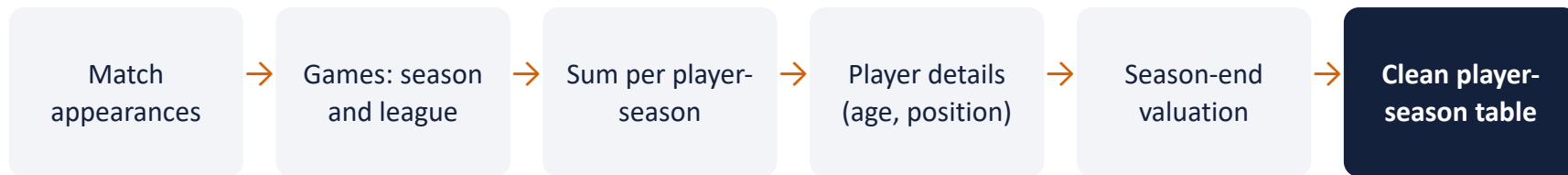
Assists

Per-90 rates

League

Market value

From 1.9 million match rows to one analytical table



450 minutes

Per-90 rates only for players with at least 450 minutes (five full matches). 1 goal in 1 minute = 90 per 90.

July 31 rule

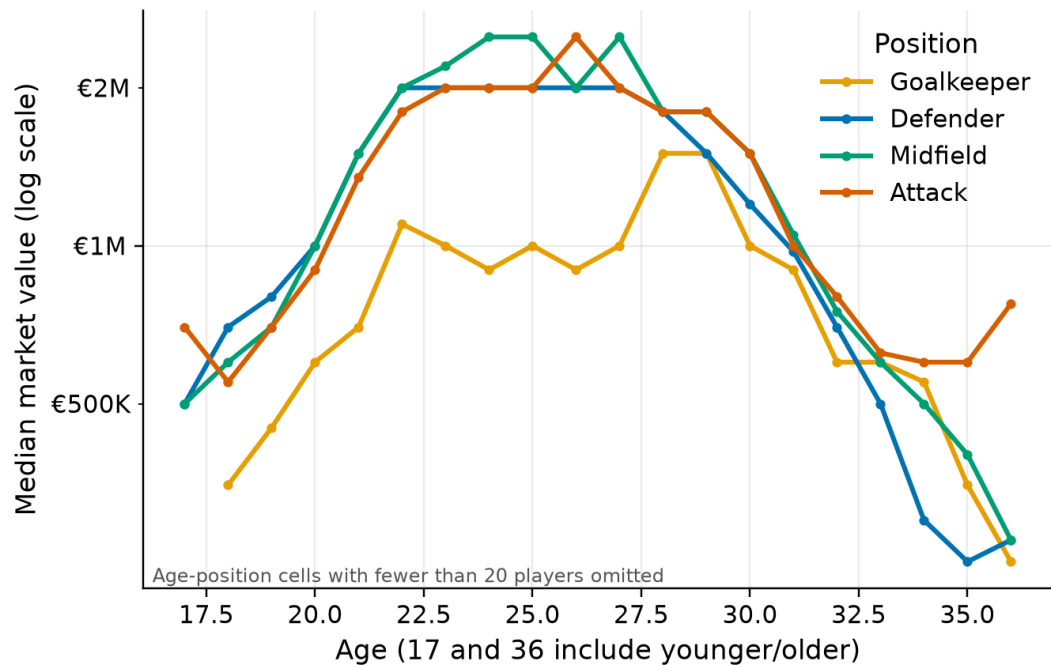
Season value = latest Transfermarkt valuation on or before July 31 after the season.

365 days

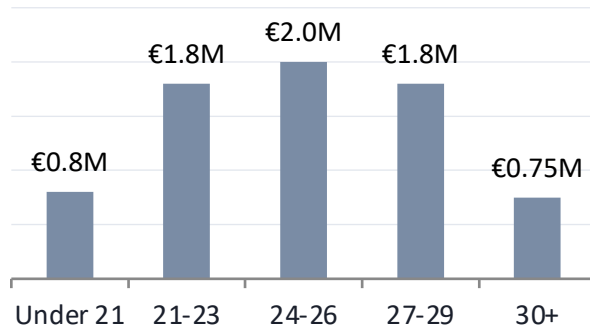
Valuations more than a year old at that date are excluded, so no stale values.

32,966 → 32,046 player-seasons: 920 excluded (2.8%) for missing position, birth date or valuation. Players changing clubs keep one row, labelled by the club where they played most.

Age matters, but not in a straight line



Median value by age group (all players)



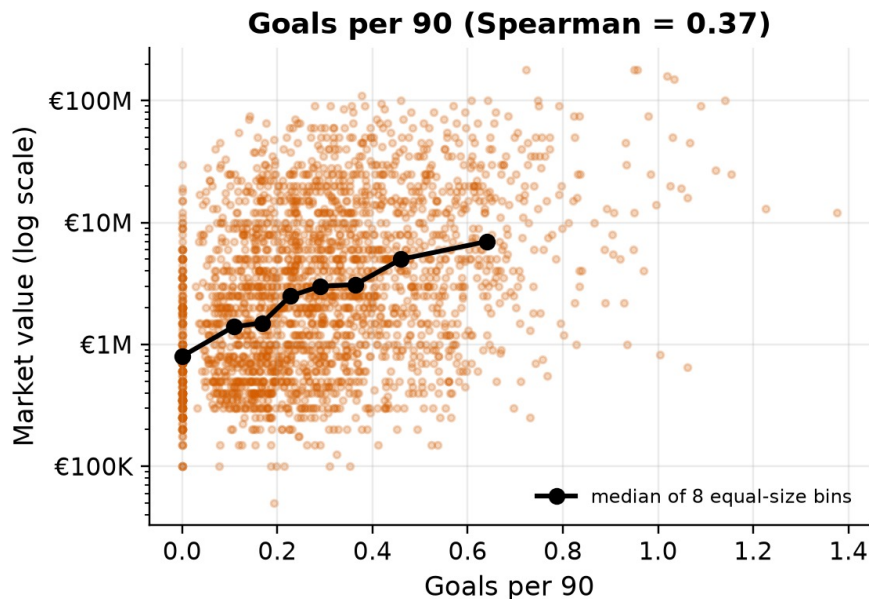
Spearman(age, value) = -0.03

Near zero, yet age clearly matters: value rises, peaks at 24–26, then falls. A single correlation hides the hump.

Median by age, one line per position; value axis is log-scaled (each step $\times 10$).

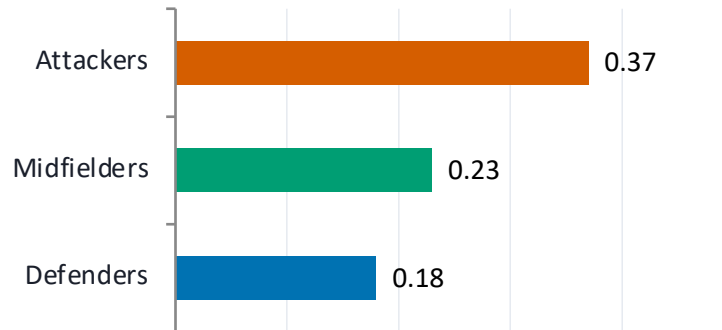
The same statistic means different things by position

Attackers with at least 450 minutes (n=6,095)



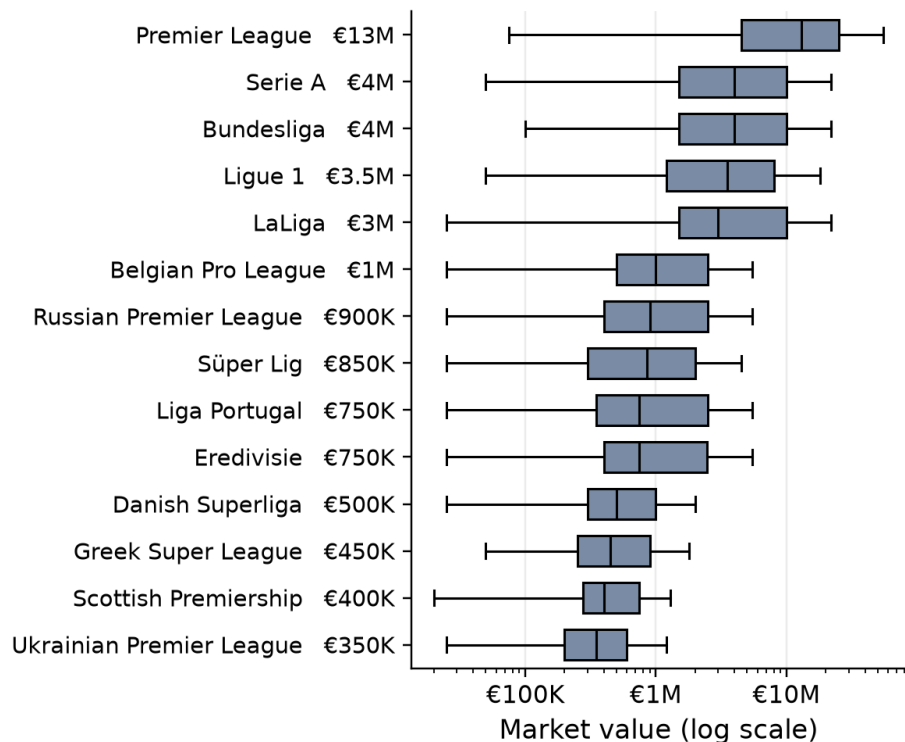
Each dot is an attacker-season; black line = medians of 8 equal-size groups.

Goals per 90 vs value (Spearman, ≥ 450 min)



Goals track value most for attackers. That does not make goals a measure of defender quality: the data has no tackles, saves or other defensive statistics.

League context shows the largest group differences



€13M

Premier League median value

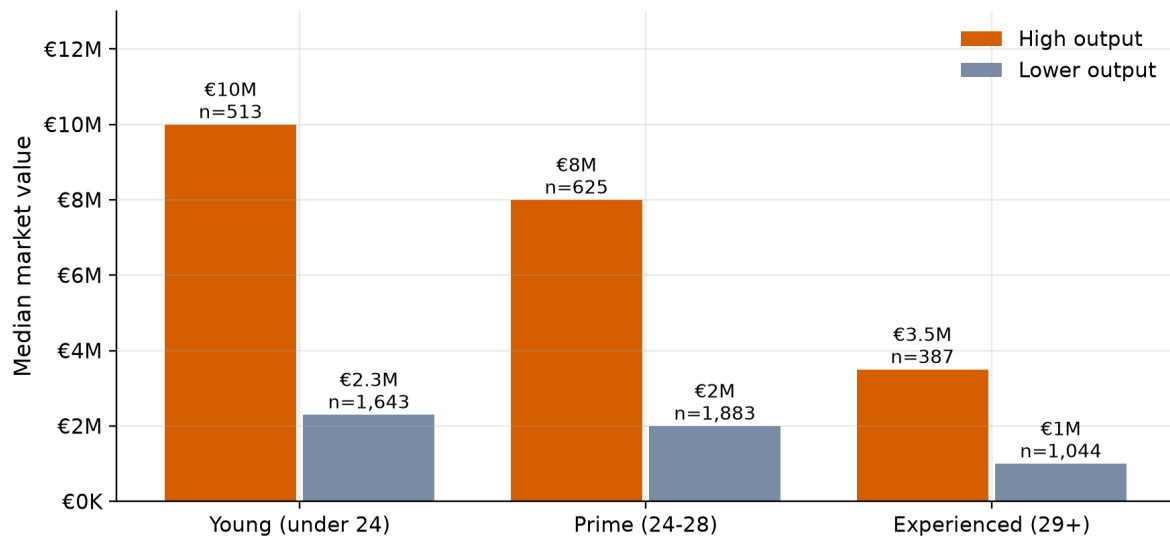
€4.5M vs €600K

top-five leagues vs the other nine (pooled medians)

Context, not cause

Leagues differ in money, clubs, player pool and exposure. About 35% of the variation in (log) value lies between leagues, vs about 1% between positions.

Young, high-output attackers: highest median values



€10M

young, high output

€8M

prime age, high output

€3.5M

experienced, high output

Attackers with ≥ 450 minutes; high output = top quarter of goals + assists per 90 (≥ 0.59). Same ordering inside the top-five leagues (€31M vs €25M) and outside them (€5M vs €4.5M). Descriptive only: potential, contracts and club level are not in the data.

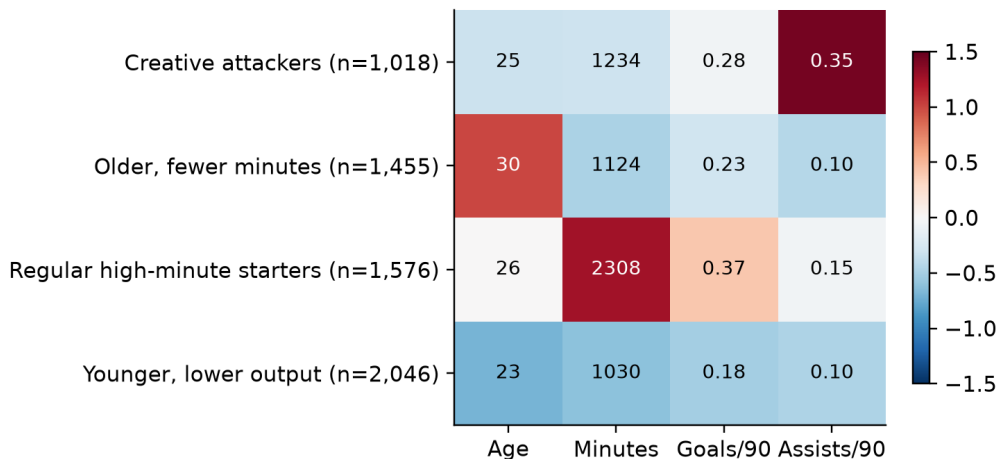
Do attackers form distinct player types?

K-means groups players with similar numbers.

- 6,095 attacker-seasons (≥ 450 min)
- Inputs: age, minutes, goals/90, assists/90
- Not used: market value, league
- $k = 2$ to 6 tested; $k = 4$ kept for readability

0.21–0.22

silhouette score at every k : weak separation. Profiles overlap; a continuum, not four true types.



€8M

Regular high-minute starters

€5M

Creative attackers

€1.5M

Younger, lower output

€1M

Older, fewer minutes

What the analysis shows

Age

Value rises into the early/mid twenties and falls after 30: a hump, not a line.

Playing time

One of the strongest simple associations (Spearman 0.51), in both directions.

Position

Performance statistics mean different things by role; goals matter most for attackers.

League

€13M Premier League median; top-five €4.5M vs others €600K.

Age × output

Young high-output attackers: highest median (€10M).

Clustering

K-means gives readable but weakly separated attacker profiles.

All associations: none of these shows that a characteristic causes a higher value.

What this project cannot show

Estimates

Transfermarkt values are estimates, not prices

No causation

Associations only; factors overlap

Missing roles

No defensive or goalkeeper statistics

No injuries

Injury history is not in the data

League and club

Context overlaps; club strength not measured

Repeated players

The same player appears in up to five seasons

Thresholds

450 min, July 31, 365 days are practical choices

Russia / Ukraine

War conditions after 2022 affect those leagues

Backup slides

For questions: exclusions, K-means details, segment definitions, league medians, playing time

Backup A: how the dataset was cleaned

Step	Rows dropped	Rows remaining
Player-seasons in scope (14 leagues, 2020–2024)		32,966
Missing position	9	32,957
Missing date of birth	5	32,952
No valuation by July 31	490	32,462
Valuation older than 365 days	416	32,046
Zero minutes	0	32,046
Final dataset	920	32,046

23,328

rows with ≥ 450 minutes (72.8%)
used for per-90 rates; others
kept for everything else

Backup B: K-means details

k	Inertia	Silhouette
2	19,125	0.219
3	16,201	0.215
4	13,850	0.207
5	12,093	0.214
6	11,075	0.209

Cluster (k = 4)	n	Age	Min	G/90	A/90	Value
Regular high-minute starters	1,576	25.9	2,308	0.37	0.15	€8M
Creative attackers	1,018	24.6	1,234	0.28	0.35	€5M
Younger, lower output	2,046	23.0	1,029	0.18	0.10	€1.5M
Older, fewer minutes	1,455	30.0	1,124	0.23	0.10	€1M

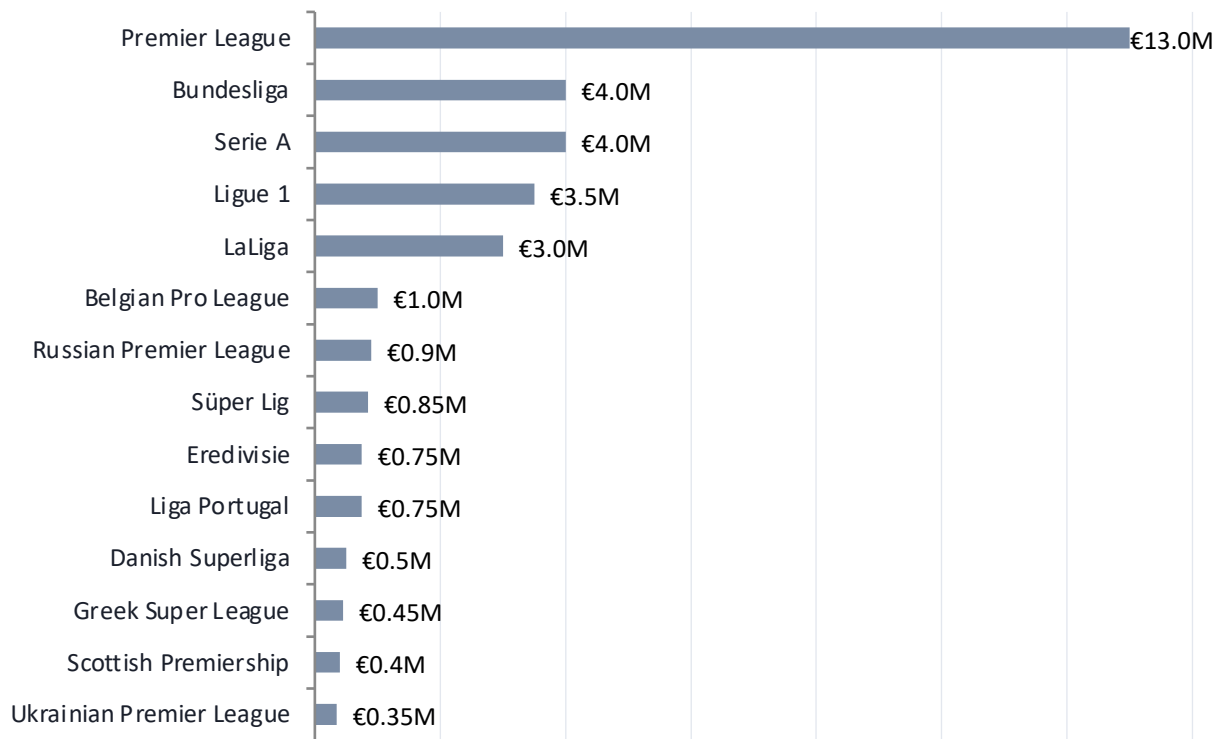
- scikit-learn KMeans, n_init = 20, random_state = 0; StandardScaler; per-90 rates capped at the 99th percentile.
- Stability (adjusted Rand index vs the seed-0 fit): minimum 0.92, median 0.97 across 10 seeds; minimum 0.89, median 0.92 on 80% subsamples.
- Clusters account for ~23% of log-value variance among these attackers.

Backup C: rule-based segments (not clusters)

Segment	Rule (applied in order)	n	G/90	A/90	Median value
High scorers	Goals/90 $\geq$ 0.40 (top quarter)	1,524	0.53	0.15	€6M
Creative attackers	Assists/90 $\geq$ 0.23, not a high scorer	1,074	0.21	0.31	€4M
Young (under 24)	Everyone else, under 24	1,235	0.19	0.11	€2M
Prime age (24-28)	Everyone else, 24 to 28	1,456	0.19	0.10	€1.8M
Experienced (29+)	Everyone else, 29 or older	806	0.20	0.10	€900K

Defined by me with explicit cut-offs from the data; easier to explain than K-means, but capture less value variation (~14% vs ~23%).

Backup D: median market value by league



€4.5M

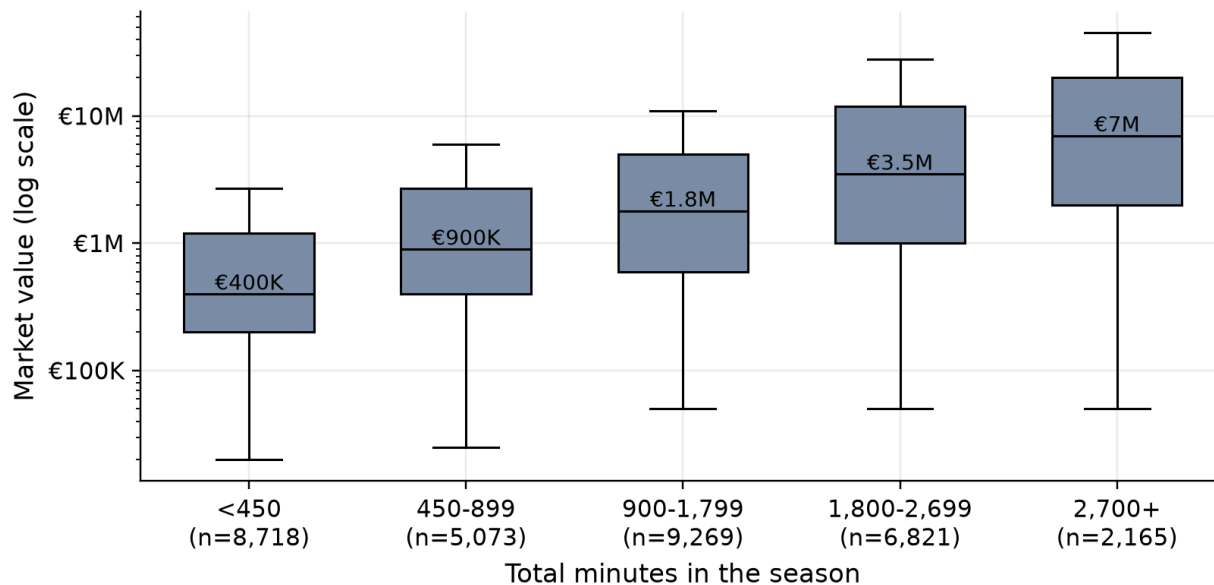
pooled median, top-five leagues

€600K

pooled median, other nine leagues

Every top-five median is €3M or more; every other league is €1M or less.

Backup E: playing time and value



0.51

Spearman, total minutes vs value (all players)

Likely runs both ways: valued players get picked, and playing raises visibility. Club level may drive both.